

Experimental investigation on the effect of interfacial properties of chemical flooding for enhanced heavy oil recovery

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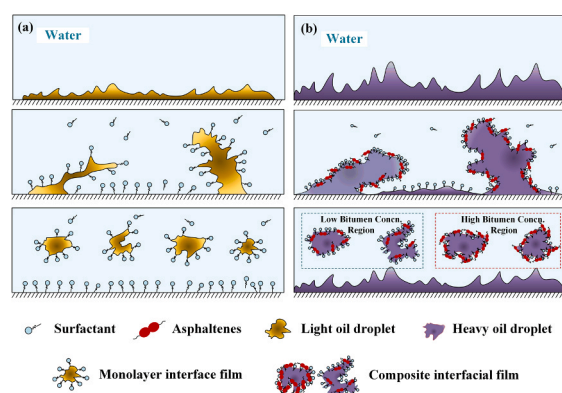
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GRAPHICAL ABSTRACT



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ABSTRACT

Chemical compound flooding has demonstrated its efficacy in enhancing light oil recovery through ultra-low interfacial tension (IFT). However, the optimization of chemical systems properties for displacing heavy oil has not been clarified. This research aims to address this problem by designing common polymer and three surfactant-polymer (SP) systems with varying IFT and interfacial viscoelasticity behaviors, and comparing their ability to recover heavy oil. The findings of core flooding tests indicate that the system with low IFT and high interfacial viscoelasticity simultaneously has a better displacement efficiency for heavy oil than other flood systems, which could enhance heavy oil recovery by 21.94%. Micro-model tests reveal the underlying mechanisms driving the enhanced macro-scale oil recovery. The emulsion formation and interface expansion facilitated by low IFT, as well as increased emulsion stability and reduced snap-off of oil droplets due to the presence of an elastic oil-water interfacial film. These phenomena are more capable of driving microscopic heterogeneous remaining oil and film remaining oil than other solution. The comparative experiment analysis shows that the difference in chemical flooding between heavy oil and light oil is caused by the discrepant distributions of

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surfactants at the oil-water interface. The light oil interface is a monolayer surfactant molecular film, the heavy components and surfactant molecules exhibit both competitive and cooperative adsorption phenomena at the oil-water interface. The occurrence of viscosity reduction and increased emulsion stability are more influential factors in the process of heavy oil displacement than those of conventional ultra-low IFT systems used for light oil recovery.

1. Introduction

In recent years, with the continuous exploitation of conventional oil resources, heavy oil, as a vital energy source, has broad development prospects, thus rendering it economically significant due to the widespread distribution and substantial reserves [1]. However, for deep heavy oil reservoirs or thin heavy oil deposits, the most widely used thermal methods are greatly restricted. Consequently, water flooding is the main development method. The recovery of conventional water flooding from heavy oil reservoirs is generally ranging between 10% and 20%, attributable to its high viscosity and high gravity. These factors result in considerable amounts of residual and remaining oil [2,3].

Over the last decades, the application of chemical flooding techniques involving polymers, surfactants, alkalis, and their combinations has become an effective method for enhancing heavy oil recovery [4,5]. Polymer flooding has gained recognition as a well-established technique for enhancing sweep efficiency. The polymer is used to increase the viscosity of the water, which effectively minimizes the mobility difference between the water and the oil, and to try to avoid fingering impacts. Initially, it was thought that polymer flooding applications were limited to reservoirs featuring oil viscosity below 200 mPa·s. However, improvement in horizontal well applications stimulated the feasibility of polymer flooding for more viscous oil reservoirs [6,7]. In comparison with polymer flooding, surfactant injection primarily aims at reducing IFT, altering wettability, and promoting emulsion formation. It is generally believed that ultra-low IFT is beneficial for its ability to reduce capillary resistance [8]. Moreover, the introduction of alkalis enhanced oil recovery (EOR) by interacting with the organic acids present in the heavy oil, thereby generating in situ surfactants [9]. Notably, the earliest research on chemical flooding was put in place for light oils, but theories elucidating the performance of chemical flooding performance in light oil reservoirs are not directly applicable to heavy oil reservoirs [10]. To shed light on the distinct behavior of chemical flooding in light and heavy oil systems, Han et al. [11] evaluated discrepant distribution patterns of sodium dodecyl sulfate (SDS) in light oil/water and heavy oil/water systems by molecular dynamics simulation techniques. They found that through hydrogen bonding, water molecules establish connections with SDS and asphaltene molecules, leading to the formation of a ternary SDS-Water-Asphaltenes (S-W-A) structure. This structure is responsible for the entrapment of SDS molecules within the heavy oil phase, whereas SDS molecules in the light oil phase spontaneously return to the oil/water interface. Furthermore, Lashkarbolooki et al. [12] revealed the equilibrium IFT of different crude oil types at varying salt concentrations, linking it to the content of asphaltene and resin fractions, as well as the aromaticity of asphaltene and resin constituents. From the above research results, it can be found that the difference in displacement characteristics between light oil and heavy oil is mainly caused by the adsorption of asphaltenes and resins at the oil-water interface. Few have provided some insights into the different phenomena of light oil and heavy oil during chemical flooding under one experimental condition.

Many studies on heavy oil have concentrated on the significance of IFT and mobility ratio. Zhang et al. [13] performed 33 sand pack flood tests, using different chemical injections to investigate how to enhance heavy oil recovery under the influence of IFT and water-phase viscosity. Their results demonstrated that the addition of the surfactant to A or AP solution can reduce IFT to an extremely low level. However, there is no positive effect on the oil recovery, so controlling the mobility of the

flooding process is more recommendable than achieving an ultra-low IFT with surfactant in heavy oil reservoirs. The laboratory core flood experiments conducted by Ding et al. [14] revealed the performances and mechanisms of SP and AP flooding for heavy oil recovery. These investigators reported that the SP system predominantly resulted in ultra-low IFT and oil-in-water (O/W) emulsification. While AP flooding prefers to form water-in-oil (W/O) emulsions. Furthermore, AP flooding exhibited significantly higher heavy oil recovery compared to SP flooding. However, the SPs were capable of maximizing heavy oil recovery by increasing the slug size using particle gels or bubble assistance. Zhou et al. [15] also obtained similar conclusions through the chemical flooding experiments in a two-dimensional physical model. They observed that enhancing heavy oil recovery mainly depends on improving the sweep efficiency after the initial waterflood. Guerrero et al. [16] employed a non-linear system (2D Model) to analyze the microscopic oil displacement mechanism in greater detail. The results indicated that the two mechanisms of surfactant flooding are oil stripping and continuous sub-fingers formation. The effect of polymer is to reduce the number of fingers and enlarge individual fingers, and the addition of surfactant does not play a significant role in heavy oil recovery. Therefore, alkalis have demonstrated a clear capability to enhance heavy oil recovery compared to polymer flooding and SP flooding. However, the low IFT exhibited by surfactants only plays a relatively minor role [17]. However, it is noteworthy that alkalis will cause reservoir damage due to the dissolution of quartz in the reservoir [18]. So it is crucial for petroleum workers to identify the surfactants that positively impact ultimate recovery during chemical-assisted recovery processes in heavy oil reservoirs. In other words, the focus should not solely be on reducing IFT for displacing heavy oil but also on determining the properties of the oil-water interface that are conducive to improving heavy oil recovery.

Viscoelasticity is also an essential property of the fluid interface, plays a significant role in determining the stability of emulsions [19]. In the context of petroleum production operations, the formation of emulsions is a common occurrence [20–23]. As heavy oil contains active components such as asphaltenes and acidic components, which obviously affect the oil-water interfacial property, the study of interfacial viscoelasticity in heavy oil has received increasing attention [24–26]. Harbottle et al. [27] established a direct correlation between the rheological properties of asphaltene films and emulsion stability, when a viscous material became predominant at the oil-water interface, drop coalescence happened in seconds. Conversely, when the film microstructure transforms into elasticity, there is no coalescence between two drops. Furthermore, Garcia-Olvera et al. [28] unearthed that the viscoelastic interface has a noticeable influence on the core flooding recovery factor by suppressing snap-off and enhancing oil coalescence during the flooding process. However, despite the critical role of surfactants in enhancing heavy oil recovery, the specific characteristics and parameters required for optimal performance, such as the role of low IFT and high viscoelasticity in residual heavy oil recovery, remain uncertain and subject to questioning due to the lack of a comprehensive mechanistic understanding.

In this paper, an assessment is conducted to evaluate the IFT, viscoelasticity, and oil-water interfacial viscosity of both single polymer and three SP binary systems. Meanwhile, oil recovery during the process of core flooding and quantitative analysis of the morphology and distribution of the microscopic remaining oil by the above chemical agents are reported. Furthermore, light oil is used for comparison throughout

this study. Through these systematic comparisons to determine the difference in chemical flooding between heavy oil and light oil. The results of this study are integrated with microscopic and macroscopic studies, thereby the dominant mechanism of enhanced heavy oil recovery is identified.

2. Materials and methods

2.1. Materials

The light oil and heavy oil are collected from the same source bed in D Oilfield, with viscosities of 22 mPa·s and 104 mPa·s at a temperature of 60 °C, respectively. The SARA contents of the two oil samples are shown in Table 1. Brine water for preparing the chemical solution and oil displacements has a TDS of 7064 mg·L⁻¹, the ionic composition of the brine water is listed in Table 2. All reagents used were of analytical-grade quality and were purchased from Sinopharm.

The first surfactant (#1) utilized was an anionic surfactant provided by D Oilfield, which is a mixed petroleum sulfonate surfactant (WPS). The second surfactant (#2) employed was a cationic surfactant known as cetyltrimethylammonium bromide (CTAB). The third surfactant (#3) utilized was a nonionic surfactant called fatty alcohol polyshiyouoxyethylene ether (AEO). The polymer employed in this work was hydrolyzed polyacrylamide (HPAM). This polymer has an average molecule weight of 30 million Daltons and exhibits a degree of hydrolysis of 25 mol%. The polymer and SP solutions were diluted using brine water.

2.2. Shear interfacial property tests

The IFT was measured using a TX5000 spinning drop interface tensiometer operating at a spinning velocity of 5000 revolutions per minute. For the assessment of shear viscoelasticity and viscosity of the water-oil interface, the Physica MCR 301 rheometer with a biconical rotor was utilized in oscillatory mode. The bicone geometry consisted of a radius of 34.14 mm, a cone angle of 5°, and a height of 2.235 mm. First, the cup was filled with the aqueous phase, and the edge of the bicone rotor was adjusted to be tangent to the water surface. Then, oil was added slowly along the rod of the rotor using a syringe until it fully covered the bicone disk, with the oil level maintained at least 1 mm above the tip of the bicone. It takes at least 8 h for the interface film of asphaltene to reach equilibrium. So the system was aged for 10 h to form the homogenized interface and shielded with the anti-volatile cap. The measurements of the modulus were conducted within the range of the corresponding linear viscoelastic region [29]. The viscosity of both bulk phases was measured at varying shear rates. While maintaining a temperature of 60 °C for all the above tests.

2.3. Core flooding experiments

The natural sandstone cores used in this study possess dimensions of 15 cm in length and 2.5 cm in diameter. The permeability of the core is between 790 mD and 881 mD, the porosity is estimated to be around 29%. The parameters of these sandstone cores are given in detail in Table 3.

Fig. 1 displays the schematic diagram of the core flooding equipment. It is mainly composed of an ISCO pump, a core holder, a pressure acquisition system, and an effluent collector maintaining the reservoir temperature at 60 °C. First, measurements of core permeability and

Table 1
The SARA contents of the two oil samples.

Oil Samples	Saturates (wt %)	Aromatics (wt %)	Resins (wt %)	Asphaltenes (wt %)
Light oil	68.47	17.32	9.36	4.85
Heavy oil	55.89	20.96	14.35	8.8

Table 2

The ionic composition of brine water.

Ion type	K ⁺ +Na ⁺	Mg ²⁺	Ca ²⁺	Cl ⁻	CO ₃	HCO ₃
Concentration (mg·L ⁻¹)	2549	49	76	3864	61	465

Table 3

The parameters of sandstone cores in core flooding experiments.

Oil type	Chemical formula		Measured properties	
			permeability (mD)	Porosity (%)
Light oil	P	0.15% P	881	27.43
	SP-#1	0.3% S+ 0.15% P	754	28.37
Heavy oil	P	0.15% P	825	29.04
	SP-#1	0.3% S+ 0.15% P	776	28.55
	SP-#2		790	28.68
	SP-#3		813	29.24

porosity were performed. Subsequently, the core is evacuated using a vacuum pump for 2 h and subsequently saturated with brine water. Afterward, crude oil was flooded at a rate of 0.1 mL·min⁻¹ until no more water is produced to establish initial oil saturation and age for 24 h. The flooding process involved three stages: water flooding terminated at a water cut of 99%, followed by chemical flooding at 1 pore volume (PV), finally, subsequent water flooding to the water cut gets to 99%. The velocity of each fluid injection was maintained at an advancement speed of 0.3 mL·min⁻¹.

2.4. Micromodel flooding studies

polydimethylsiloxane (PDMS) was selected as the material for micromodels, which performed oil-wet characteristics, exhibiting a contact angle of approximately 120°. The micromodels are designed with pore dimensions measuring 30 µm in-depth, 6 mm in length, and 4 mm in width. The microfluidic setup as depicted in Fig. 2, displacement energy is supplied by a microflow syringe pump (Harvard PHD ULTRA). Images are acquired through a stereoscopic microscope (ZEISS Stereo Discovery V12) equipped with a camera (SONY A7S2), boasting an image resolution of 4240 × 2832 pixels.

The micromodels were initially saturated with oil samples and aged for 0.5 h. Thereafter, water and either polymer or SP solutions were sequentially injected into the micromodels at a velocity ranging from 3 to 5 µL·min⁻¹. Throughout the injection process, the displacement process is monitored through a camera. Each displacement stage lasts sufficiently long to ensure the validity of pore-throat mobilization. The image analysis method proposed by Gao et al. was adopted to analyze and calculate the content of different remaining oil types for each image [30,31].

3. Results and discussion

3.1. Shear interfacial property tests

In this section, the IFT, shear viscoelasticity, and viscosity between heavy oil and various chemical systems were tested to provide powerful evidence of the structure of the interfacial layer, which is significant to characterize interfacial properties.

3.1.1. IFT tests

The value of IFT presented in Fig. 3 exhibits a decreasing trend over time. The experiments are conducted for 1 h to reach equilibrium IFT within the range of 10⁻¹–10⁻² mN·m⁻¹. Notably, the attainment of a stable level of IFT takes longer for heavy oil-water systems. All three surfactants demonstrated clear efficacy in reducing the oil-water IFT. Surfactant #1 proved to be the most efficient in reducing the IFT, whether it was at the interface of heavy oil/SP solution or light oil/SP

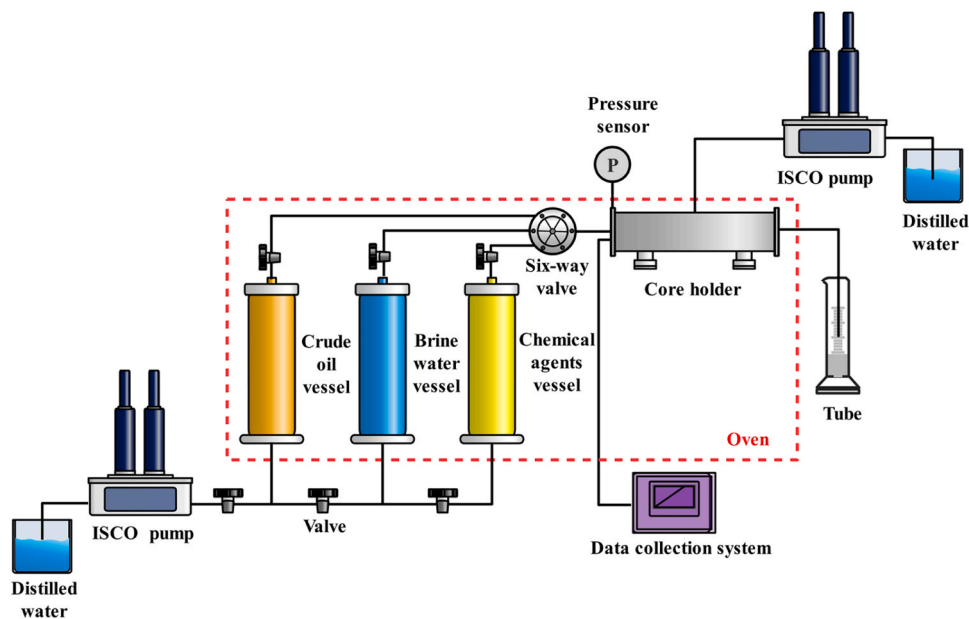


Fig. 1. The schematic diagram of core flooding tests.

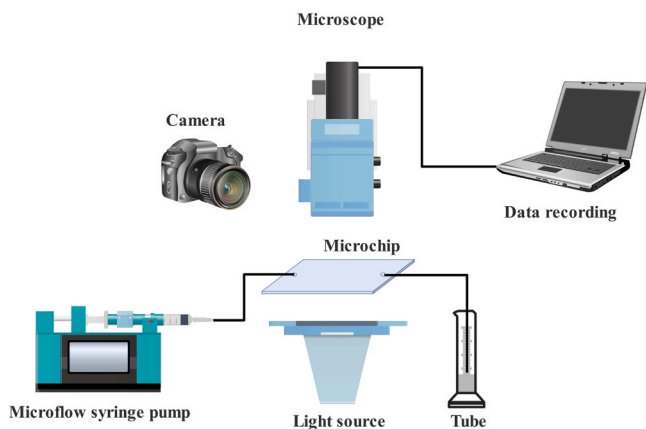


Fig. 2. The schematic diagram of Micromodel flooding tests.

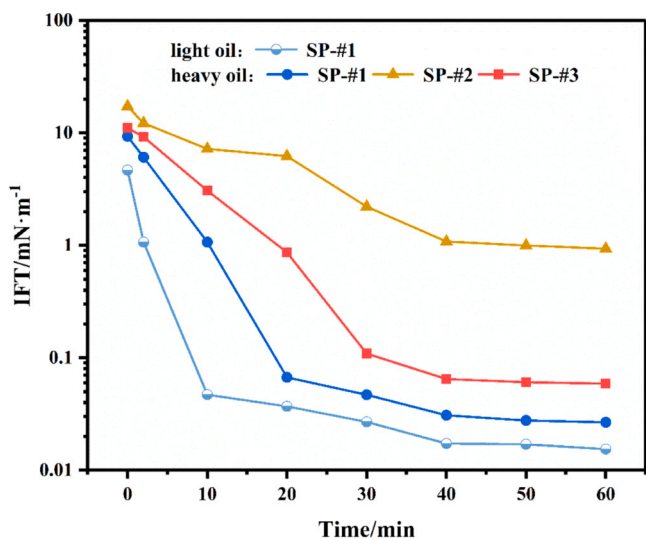


Fig. 3. Dynamic IFT behavior between oil and different displacement solutions.

solution. The IFT of heavy oil/SP-#1 solution ($0.0267 \text{ mN}\cdot\text{m}^{-1}$) was higher than that of the light oil system ($0.0154 \text{ mN}\cdot\text{m}^{-1}$). Furthermore, the IFT value of heavy oil and SP-#2 is marked higher, reaching $0.934 \text{ mN}\cdot\text{m}^{-1}$. On the other hand, the SP system composed of SP-#3 achieved a significant reduction, reaching $0.059 \text{ mN}\cdot\text{m}^{-1}$.

3.1.2. Shear viscoelastic tests

The interfacial shear viscoelasticity plays a crucial role in evaluating the interaction between the displacing and displaced phases during the oil displacement process, reflecting the long-term stability of the interfacial film [32]. The rheological parameters, including the storage modulus and loss modulus, serve as indicators of the elasticity and viscosity of the interfacial film, respectively. The interfacial viscoelasticity for three distinct chemical systems and heavy oil was measured, as shown in Fig. 4. When the interfacial film is subject to shear, the active molecules at the interface align and form a film. The intensity of the interfacial film between HPAM and heavy oil is comparatively weak. Within the shear rate range of $0.01\text{--}0.1 \text{ rad}\cdot\text{s}^{-1}$, elastic modulus remains

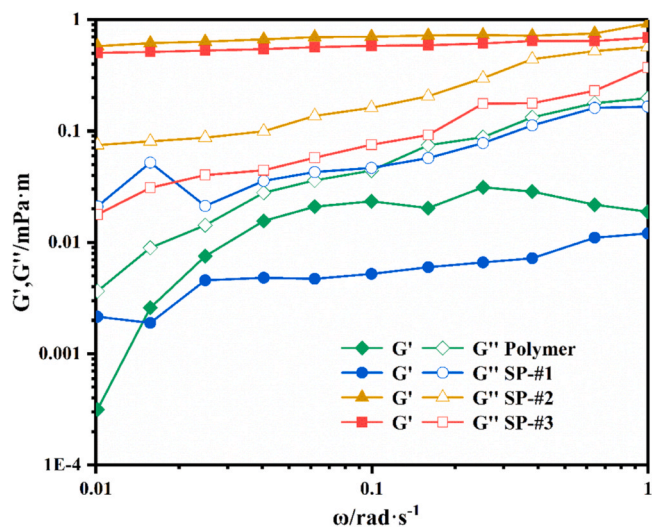


Fig. 4. Comparison of viscoelasticity for heavy oil with different chemical solutions.

basically unchanged with increasing shear rates. This behavior can be attributed to the weak interaction between HPAM and crude oil, as the former lacks active components. Conversely, under higher shear stresses ($> 0.5 \text{ rad}\cdot\text{s}^{-1}$), the molecule order at the oil-water interface becomes disrupted, leading to a reduction in the elastic modulus as the interfacial film structure is destroyed. In addition, the interface viscoelasticity may be tuned when adding a surfactant. The G'' between heavy oil and the SP-#1 system extremely surpasses that of G' , indicating the prevalence of viscous deformation. However, an elastic behavior between the SP-#2 and SP-#3 systems and heavy oil is observed throughout the entire frequency range, as evidenced by $G' > G''$. This observation can be attributed to the strong hydrogen bonding between the molecules within the interfacial membrane.

3.1.3. Shear viscosity tests

The interfacial shear viscosity is a fundamental parameter that characterizes the molecular interactions at the oil-water interface. It depends on the arrangement tightness and interaction strength among neighboring film-forming molecules, as well as the presence of organized structural formations. At the oil-water interface, asphaltenes containing active components can organize cross-link networks, which are considered effective modifiers for altering interfacial properties in a mechanical manner [33,34]. However, the presence of different displacement systems adsorbed to the interface undoubtedly leads to competitive adsorption with asphaltenes, which can soften the elastic interface between oil and water and influence the rheology of the interfacial film [35]. Fig. 5 shows the interfacial shear viscosity between heavy oil and various chemical systems. The results indicate that the conventional polymer exhibit weak interfacial interaction with heavy oil due to the absence of active and functional groups. However, the addition of surfactants to the polymer substantially enhances the interfacial shear viscosity. Because the functional groups of the surfactants are closely arranged on the interface, which forms strong hydrogen bonds with the active components of the heavy oil. Surfactants are known to accumulate at the fluid-fluid interface, preventing the interaction between the polar components in the heavy oil and the water ions. At the same time, surfactants may also stabilize asphaltenes in the bulk phase of oil, preventing them from stacking and forming alternative structures at the interface. Comparing the interfacial shear viscosity of SP-#1 system with that of heavy oil, the oil-water interfaces of the SP-#2 and SP-#3 exhibit higher values, resulting from the stronger intermolecular interactions facilitated by #2 and #3 surfactants at the oil-water interface. Moreover, the interfacial shear viscosity of SP-#1 tends to

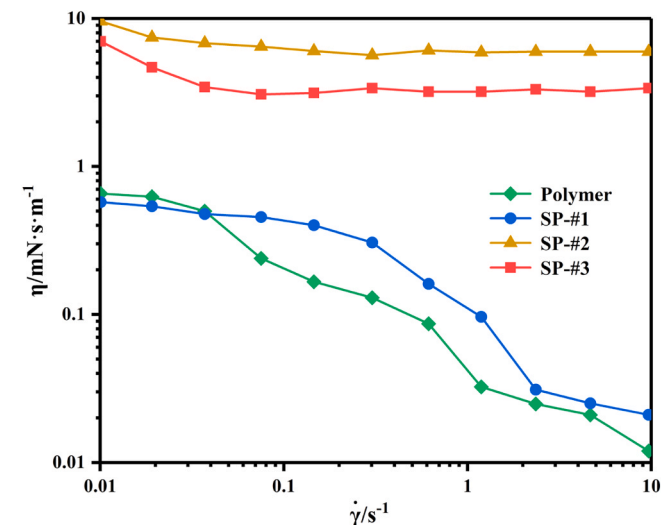


Fig. 5. Comparison of shear viscosity for heavy oil with different chemical solutions.

decrease as the shear rate exceeds 0.5 s^{-1} , indicating a weaker strength of the interfacial film under higher shear rates.

In terms of oil-water interfacial properties, the three SP binary systems exhibit distinct characteristics. The SP-#1 system shows low IFT and viscous interfacial shear viscoelasticity. On the contrary, the SP-#2 system demonstrates high IFT and a substantial interfacial shear elastic modulus, showing elasticity. While the SP-#3 system exhibits both low oil-water IFT and undergoes elastic deformation. As a result, a comparative experiment can be conducted between SP-#1 and SP-#3, which have similar IFT but different interfacial shear viscoelasticity. Likewise, a comparative experiment can be performed between SP-#2 and SP-#3, which have different IFTs but similar interfacial shear elastic modulus.

3.2. Core flooding experiments

A series of six core flooding experiments were carried out to investigate the impact of polymer and surfactant on improving oil recovery in the context of heavy oil versus light oil. The oil recovery of each stage obtained from the core flooding tests is summarized in Table 4.

3.2.1. Core flooding experiments of light oil

The comparative analysis of oil recovery between two types of chemical flooding is presented in Fig. 6(a). The water flooding recovery of conventional oil was about 47%. However, after the chemical flooding and the subsequent water flooding, both the final oil recovery for polymer flooding and SP flooding can be reached 65.51% and 74.1%, respectively. The mixture of polymer and surfactant demonstrated the highest oil recovery, which increased by 26.88% after water flooding, leading to a significant increment in cumulative oil recovery. The addition of surfactant can reduce IFT, resulting in a significant increase in total recovery efficiency. Notably, the lower the IFT between the displacing phase and the displaced phase, the more obvious the effects on light oil recovery.

As shown in Fig. 6(b), injection pressure during the displacement process was analyzed for EOR mechanisms associated with the chemical flooding of light oil. It can be seen that the injection pressure exhibited an increase upon the injection of different chemical systems. However, the injection pressure rise ranges are different. polymer flooding exhibited the highest peak pressure value, reaching approximately 0.866 MPa, surpassing that of SP flooding, whose peak pressure is about 0.526 MPa. This discrepancy can be attributed to multiple factors. On the one hand, the presence of surfactants led to a reduction in the viscosity of the displacement phase. On the other hand, the low IFT and emulsifying ability facilitate oil mobility, resulting in greater oil production. These factors collectively played a significant role in the efficacy of SP flooding.

Table 4

Summary of the results of core flooding experiments.

Oil type	Chemical formula	Oil recovery (%)			
		Water flooding	Chemical flooding	Subsequent water flooding	Total
Light oil	P	47.06	17.47	0.98	65.51
	SP-#1	47.21	25.43	1.45	74.1
	S+ 0.15% P				
Heavy oil	P	27.89	13.01	0.73	41.63
	SP-#1	28.95	14.96	1.17	45.08
	S+ 0.15% P				
	SP-#2	28.75	15.93	0.76	45.44
	SP-#3	29.43	20.62	1.32	51.37

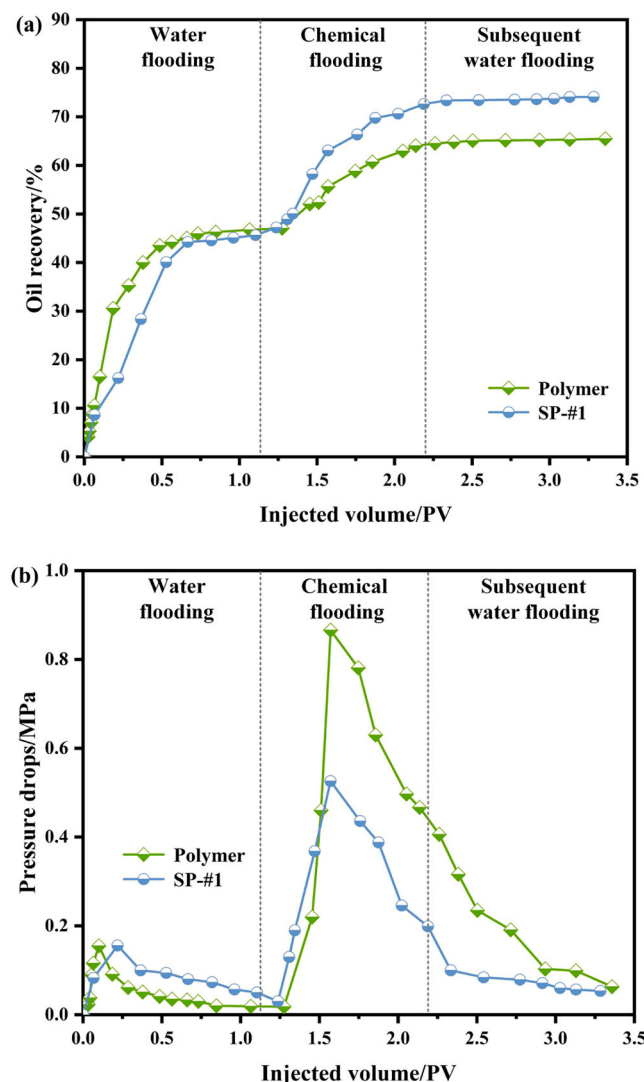


Fig. 6. (a) Oil recovery and (b) pressure drop as a function of injection volume with different chemical solutions.

3.2.2. Core flooding experiments of heavy oil

In order to elucidate the effect of various interfacial properties in SP flooding systems, core flooding experiments for pure polymers as well as three kinds of SP binary systems were conducted. When compared to conventional light oil tests conducted during polymer flooding, as depicted in Fig. 7(a), there is no apparent increment in cumulative oil recovery for the SP-#1 flood test. The result suggests that the benefits derived from a reduction in IFT are insufficient to compensate for the reduction in viscosity, resulting in a loss in oil recovery. In other words, the contribution of low IFT reduction may not be the primary mechanism responsible for enhancing heavy oil recovery. The incorporation of the #2 surfactant into the polymer solution resulted in an increase of 15.93% as compared to the conventional brine flooding, which is merely 1% higher than the SP-#1 system. Hence, it is clear that the high viscoelasticity property of the displacement phase does not significantly contribute to the efficiency of heavy oil recovery. Notably, the synergies of low IFT and high viscoelasticity in the SP-#3 system improved the oil recovery to a greater extent than polymer or other SP floods with sole function. Ultimately, by injecting SP-#3 system solution, heavy oil recovery can increase to 51.37% after subsequent water flooding.

The pressure drop curves obtained from the four flooding experiments are recorded in Fig. 7(b). Polymer flooding also showed a high displacement pressure similar to that observed with light oil. Upon the

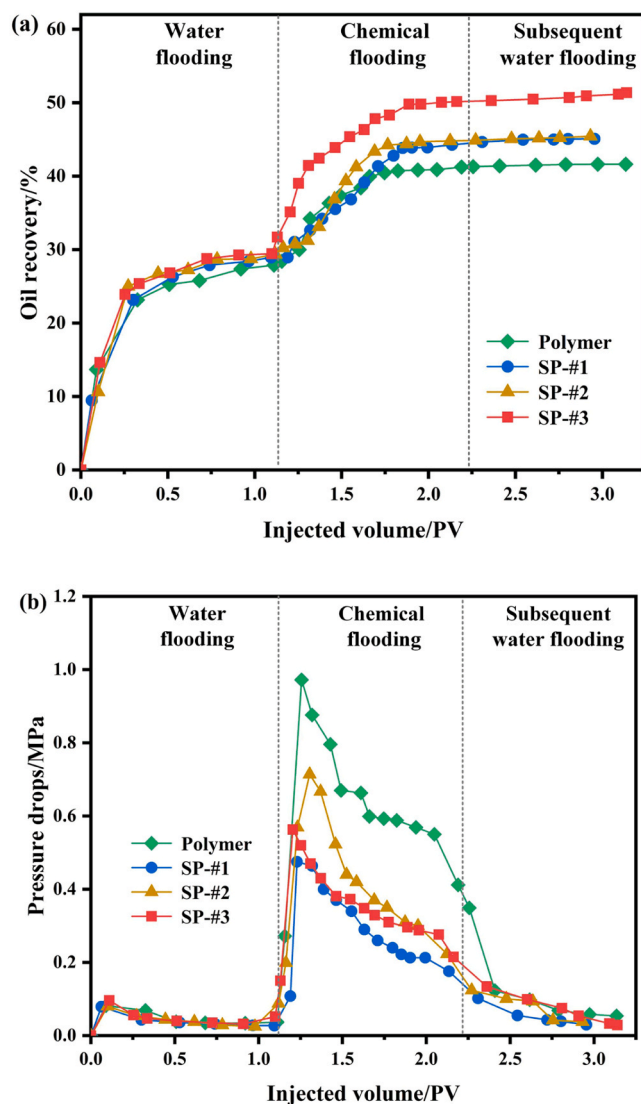


Fig. 7. (a) Oil recovery and (b) pressure drop as a function of injection volume with different chemical solutions.

addition of #1, #2, and #3 surfactants to the polymer solution individually, the highest injection pressure decreased from 0.972 MPa to 0.475 MPa, 0.714 MPa, and 0.563 MPa, respectively. This reduction in injection pressure can be explained by the facile formation of O/W emulsion resulting from surfactants possessing low IFT. Such emulsion is of benefit to the mobility of heavy oil, resulting in lower peak pressure and shifting to a later time. Additionally, Between SP-#1 (low IFT) and SP-#3 (high viscoelasticity), the pressure peak of SP-#3 is located. Consequently, an optimal range may exist where both effects are maximized, resulting in a significant increase in oil recovery.

3.3. Micromodel flooding studies

To investigate the potential enhancement of heavy oil recovery through chemical flooding, it is necessary to explain the observed phenomenon that happened in the above core displacement test. Therefore, a series of six micromodel flooding tests were conducted, and the experimental scheme was identical to the core flooding experiment in the previous section.

Based on their morphological characteristics (elliptic and narrow) and connectivity features (interaction with pores or matrixes and contact quantity), the method we used divides the remaining oil and

residual oil into the following four distinct types: pore-throat residual oil, film remaining oil, hydrodynamic residual oil, and microscopic heterogeneous remaining oil. Pore-throat residual oil typically exhibited slit-shaped and columnar-shaped structures, whose formation mechanism is that capillary force keeps it stranded in pores and throats. The reason for the emergence of film remaining oil is the adsorption of oil on the surface of reservoir rocks in either a continuous or discontinuous manner, facilitated by viscous forces. The residual oil emerged in regions that were inaccessible to the displacing phase, such as dead-ends and blind-ends, is called hydrodynamic residual oil. Lastly, the microscopic heterogeneous remaining oil is left behind in relatively low permeability areas caused by the reservoir heterogeneity, most of it was continuously distributed by clustered and intergranular adsorption.

3.3.1. Micromodel flooding studies of light oil

The altered images in Fig. 8 are captured at the end of the different stages of micromodel experiments, presenting the distribution of residual oil types following different processes. Fig. 8(a) shows the initial oil-saturated state. As depicted in Fig. 8(b), the advance path of the water phase encounters the influence of microscopic heterogeneity within the pores and throat structures of the flow space, flowing channels with reduced capillary resistance and establishing a dominant seepage channel.

As exhibited in Fig. 8(c), the polymer solution was injected into the microscopic model subsequent to water flooding. A substantial driving force forms during the start-up process of the remaining oil due to its high viscosity, so that the oil droplets could be peeled off from the surface of the pore throat. What's more, it effectively obstructed the formation of fingering zones, thereby permitting the solution to enter the non-main flow channel. The application of HPAM for enhanced oil recovery is mainly by increasing the displacement area.

The sweeping area resulting from SP flooding in Fig. 8(d) is significantly larger than that of polymer flooding in Fig. 8(c). What's more, it demonstrates enhanced oil recovery capabilities, leading to a notable improvement in displacement efficiency. This effect can be primarily attributed to the low IFT between the surfactant and oil, consequently generating a higher capillary number. This favorable condition

facilitates the expulsion of residual oil between particles through the action of capillary force, thus reducing residual oil saturation.

3.3.2. Micromodel flooding studies of heavy oil

The microscopic images presented in Fig. 9 depict the residual oil distribution resulting from waterflooding and chemical flooding (polymer flooding or SP flooding). It can be seen in Fig. 9(a) that a substantial quantity of heavy oil remained inside the model pores after long-term water flooding. Viscous fingering occurred obviously from the injection end to the production end. This phenomenon is attributed to the pronounced viscosity disparity between heavy oil and water. The subsequent water predominantly follows the primary flow channel, where a relatively thick oil film adheres to the pore surface. Consequently, a great deal of microscopic heterogeneous remaining oil is mainly distributed on both sides of the main flow line [36].

Polymers exhibit similar displacement mechanisms, whether in heavy oil or light oil. As shown in Fig. 9(b), when the viscoelastic polymer solution flows in the channel, first of all, it pushes the residual oil unaffected by water flooding. This displacement mainly has an obvious effect on microscopic heterogeneous remaining oil. In traditional capillary theory, the impact of oil viscosity is typically ignored. But as the oil viscosity rises, the influence of capillary number diminishes for the same IFT, flow rate, and displacement fluid viscosity. This reveals that viscous forces prevent the mobilization of heavy oil at the pore scale [37]. Therefore, the mobilization difficulty of residual oil increases during the displacement process.

It can be seen from Fig. 10(a) that although the oil saturation in the microscopic model decreased subsequent to displacement by SP-#1, a substantial volume of residual oil existed in various states in the microscopic visualization model. SP-#1 is a composite system with low IFT, that facilitates the deformation and separation of the remaining oil in porous media. As shown in 10(b), a large number of "oil silks" engenders in the direction of displacement of the binary system. These "oil silks" continue to elongate along the flow channel until it breaks. However, the oil recovery is predominantly governed by viscous flow, characterized by the value of $G' < G''$ between SP-#1 and heavy oil. It means that even if the oil-water interface expands easily under the

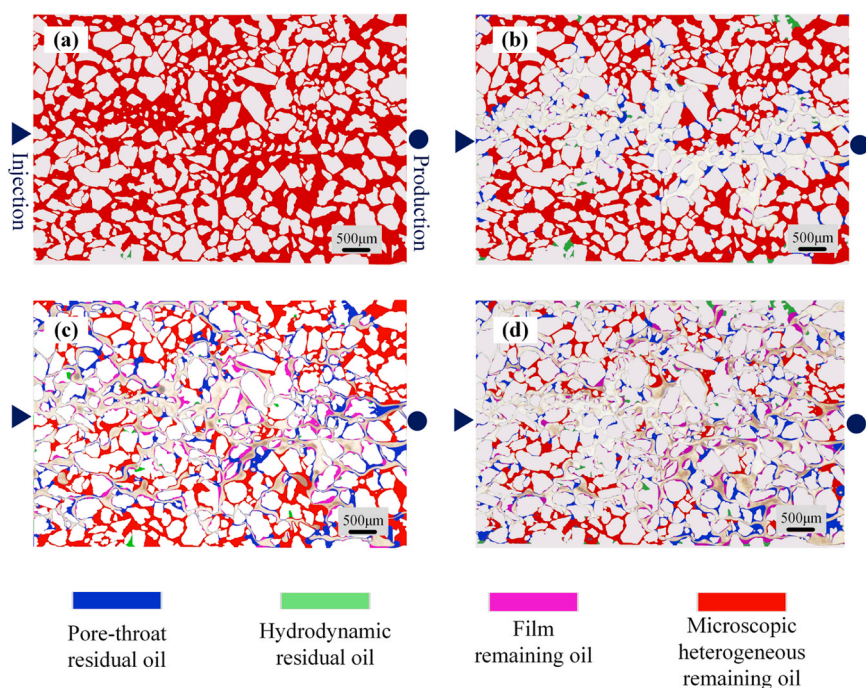


Fig. 8. The remaining oil classification images of different displacement methods using image recognition techniques: (a) initial oil-saturated state, (b) water flooding, (c) polymer flooding, and (d) SP-#1 flooding.

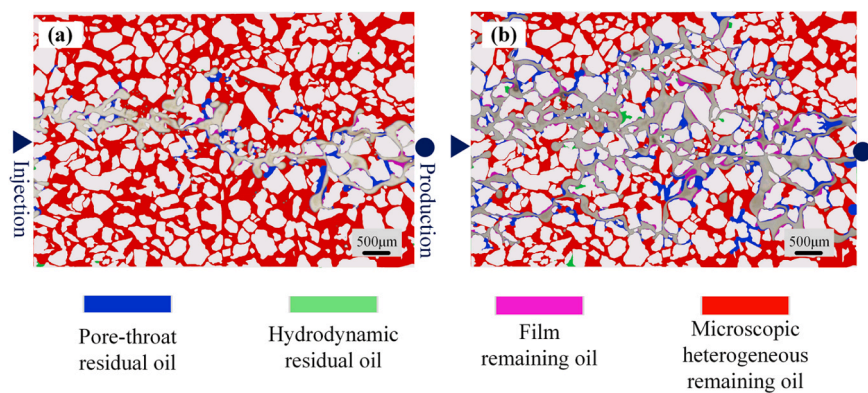


Fig. 9. The remaining oil classification images of (a) water flooding and (b) polymer flooding using image recognition techniques.

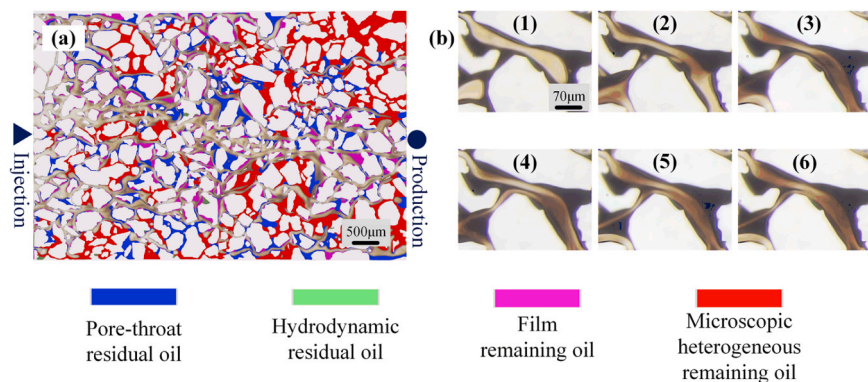


Fig. 10. (a) The remaining oil classification images of SP-#1 using image recognition techniques. (b) the creation of "oil silks" and their subsequent entrapment in the pore medium.

function of low IFT and the formation of minimal emulsified oil droplets, these droplets will quickly coalesce and transform into strips due to their poor stability [38]. As a result, A part of the mobilized remaining oil is entrapped again within the pore medium during the migration process. Although the oil-water IFT drops to ultra-low and the capillary number increases, the residual oil in the pores for its high viscous force exerted by heavy oil still difficult to be driven solely through capillary forces. Consequently, only a small part of the remaining oil is released, resulting in a low oil displacement efficiency, which sharply contrasts with the behavior observed in light oil.

As depicted in Fig. 11(a) and (b), the binary flooding process of SP-#2 demonstrates that a thick oil film formed during waterflooding

gradually stripped from the surface of the microscopic model from long to short, from thick to thin, through the combined effects of "stretch" and "drag". The interfacial film is more prone to stretching and deformation in contrast with polymers, primarily due to the presence of surfactant active groups. Thus, the previously immobile oil adhered to the wall of the main channel becomes mobile oil, under the push of the activated remaining oil and the elastic binary solution. The extent of residual oil displacement resulting from the "stretch" and "drag" effects increases with the viscoelasticity of the SP system. The elevated interfacial shear viscoelasticity profoundly augments the exfoliation and carrying capabilities of the displacement system, which is the main cause of the reduction in the remaining oil saturation [39]. However, only a limited

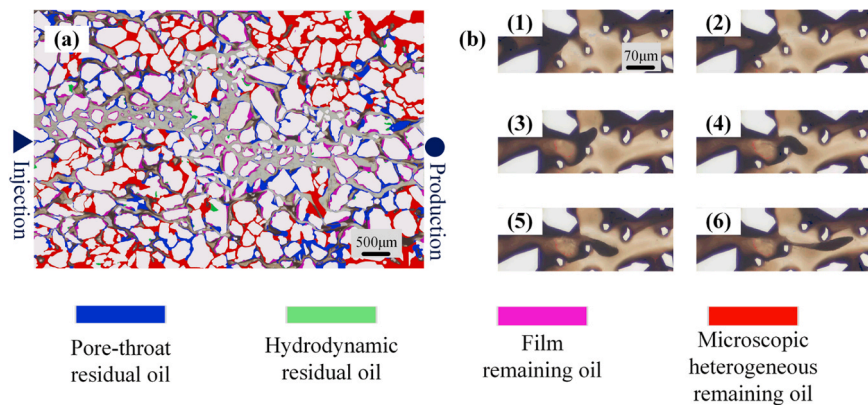


Fig. 11. (a) The remaining oil classification images of SP-#2 using image recognition techniques. (b) Oil-carrying capacity due to the elevated interfacial shear viscoelasticity.

number of emulsified oil droplets are formed during the displacement process, regrettably, this property restricts the SP solution from entering more pores and expanding the swept volume, the improvement proves insufficient in increasing oil recovery.

Fig. 12(a) shows the result for the #3 poly-surface binary system at the pore level. the low IFT leads to the formation of an O/W emulsion consisting of tiny droplets. It is noteworthy that this O/W emulsion is different from the "oil silk" created by SP-#2 system, the flowing ability of O/W emulsion assumes a more prominent role in the case of heavy oil for its low viscous resistance. It can be seen in Fig. 12(b) that as the O/W emulsion migrates forward stably, it comes into contact and converges with other emulsions, so that the heavy oil tends to migrate in the form of oil strips and advance further. Consequently, more and more unactivated residual oil starts to flow, and Residual oil saturation is also getting lower and lower. What's more, the remaining oil will gradually generate a tipnitially, then be continuously absorbed, and elongated, followed by fracture, as presented in Fig. 12(c). This is attributed to a strongly interfacial film formed by the active group of the SP-#3 system along with crude oil asphaltenes. When the solution and the residual oil flow relative to each other, it is equivalent to exerting a shearing action on the interface, effectively "dragging" the residual oil through the high viscoelasticity of the interfacial film [40]. These "dragging" forces play a vital role in enhancing washing efficiency, particularly concerning pore-throat and hydrodynamic residual oil. Therefore, on the one hand, the SP-#3 system exhibits the function of a polymer, enabling interfacial interaction through the intertwining between the crude oil and the viscoelastic effect, effectively "dragging" the crude oil out. On the other hand, #3 surfactant is capable of reducing heavy oil viscosity, the low IFT facilitates the O/W emulsion of small droplets easy to flow without adhesion to pore surfaces. The improved flow behavior of the small droplet emulsion enhances oil displacement efficiency.

3.3.3. Quantitative analysis of different types of remaining oil

Through employing intelligent identification and conducting statistical analysis of the images captured at different flooding stages, the oil saturation of various types of microscopic remaining oil can be depicted as a function of the recovery degree of different chemical fluids. As shown in Fig. 10, after waterflooding, the predominant types are

microscopic heterogeneous remaining oil and residual pore-throat oil. However, With the polymer injection volume increase, there is a significant reduction in the microscopic heterogeneous remaining oil. Specifically, heavy oil decreased from 71.38% to 42.6%, while light oil decreased from 59.69% to 22.1%. At the same time, due to the heterogeneity of the microscopic model, the partially activated microscopic heterogeneous remaining oil converts into pore-throat residual oil, so the volume of the latter increased by 9.62% for light oil and 5.5% for heavy oil. However, the quantities of hydrodynamic residual oil and filmy oil remain essentially unchanged, indicating that conventional polymers are ineffective in displacing stubborn, patchy residual oil. In contrast, the implementation of a low IFT system SP-#1 has a positive effect on various remaining oils, subsequent to water flooding, particularly in the case of light oil. The swept volume increases significantly and activates a large amount of residual oil remaining in the non-main channel of the microscopic model. Compared with polymer flooding, the oil saturation of microscopically heterogeneous remaining oil decreased by 7.32%. Simultaneously, the effect of low IFT leads to reductions of 0.82%, 0.83%, and 0.46% in pore-throat residual oil, filmy oil, and hydrodynamic retained oil, respectively. Although there is a slight decrease in oil saturation, SP-#1 flooding effectively removes a significant amount of various types of remaining oil in terms of overall quantity.

During the SP flooding stage of heavy oil, the volume of different remaining oils shows distinctly different trends as injected into different chemical systems. Upon injection of SP-#1 solution into the micro model, the microscopic heterogeneous remaining oil that exists in the main channel is emulsified and dispersed. The film remaining oil can be produced due to the displacement of the microscopic heterogeneous remaining oil, which leads to an increase instead of a decline. Finally, the microscopic heterogeneous remaining oil accounted for 39.55% of the total oil volume, 11.83% of the pore-throat residual oil, 2.45% of the hydrodynamic residual oil, and 5.09% of the filmy oil. On the contrary, SP-#2 solution shows poor emulsification and erosion, and the content of microscopic heterogeneous remaining oil declined slowly. The increments observed in pore-throat residual oil and filmy oil are minimal, standing at 2.41% and 2.94%, respectively. which is the smallest increase among all displacement systems. This finding suggests that the

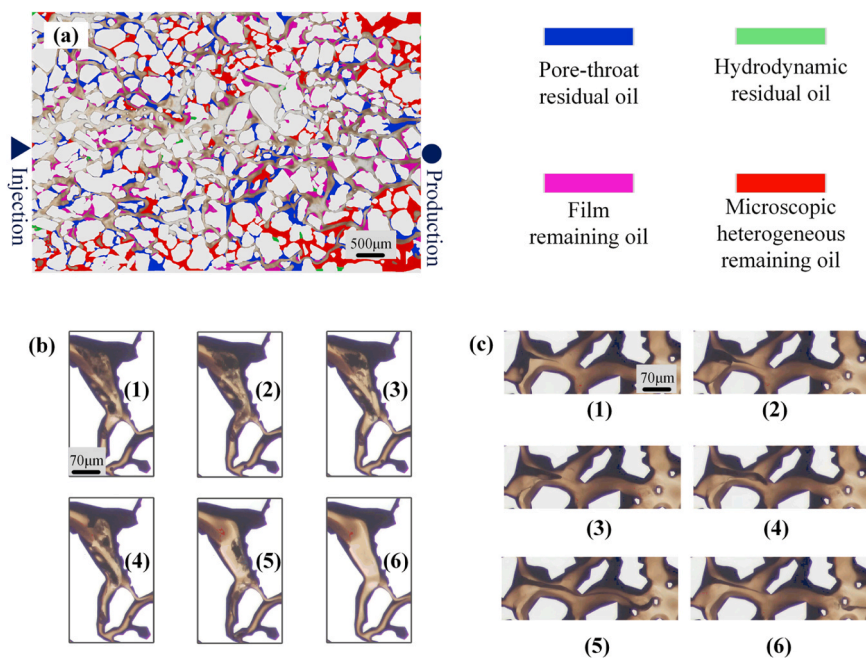


Fig. 12. (a) The remaining oil classification images of SP-#3 using image recognition techniques. (b) The O/W emulsion converges with each other and migrates further. (c) The remaining oil is absorbed, elongated, and then "dragged" away by the high viscoelasticity of the interfacial membrane.

inertial force of SP-#2 surpasses the viscous and capillary forces, due to the formation of cross-link networks at the oil-water interface, thereby enhancing the carrying capacity of displacement systems. Comparatively, SP-#3 demonstrates higher oil recovery rates, surpassing those of SP-#1 and SP-#2 by approximately 6.14% and 6.36%, respectively. The superior residual oil recovery of SP-#3 can be attributed to its high viscoelasticity. This characteristic combines with decreasing the oil-water mobility ratio significantly reduced the contents of the microscopic heterogeneous remaining oil by 33.78%. Meanwhile, the micro model retains 10.16% of the pore-throat residual oil, 4.78% of filmy oil, and 1.79% of hydrodynamic residual oil. Fig. 13.

3.3.4. EOR mechanism difference between heavy oil and light oil

It can be seen in Fig. 14(a) that the stability of the light oil-surfactant interface is primarily due to the presence of a uniform monolayer surfactant molecular film. These surfactant molecules exhibit a relatively uniform arrangement and tend to adsorb in a vertical orientation at the oil-water interface. Essentially, cross-linking or aggregation phenomena are essentially absent. Therefore, a dense interfacial film and low IFT are formed, and as a consequence, the oil recovery is enhanced during chemical flooding.

The oil-water interface of heavy oil exhibits a highly complicated nature [41]. Asphaltene molecules and surfactant molecules engage in

both competitive and cooperative adsorption phenomena at the oil-water interface. The dominant mechanism of adsorption depends on the interfacial activity of the surfactant molecule. If the surfactant molecule possesses high interfacial activity, competitive adsorption becomes prominent. Eventually, the asphaltene molecules are squeezed out, allowing the surfactant molecule to adsorb onto the oil-water interface, manifested by low IFT. Conversely, if cooperative adsorption dominates, the surfactant molecules cross-link with the asphaltene molecules to form a dense and firm interfacial film system, exhibiting high viscoelasticity. When both types of adsorption exist simultaneously, a two-layer interfacial model was proposed by Wu and Czarnecki [42] to describe the arrangement of asphaltenes and surface-active species at the oil-water interface. As shown in Fig. 14(b), based on this model, in the region of low bitumen concentration, the interface is occupied by surfactant molecules. The attraction between the substance and the top layer of asphaltenes, potentially facilitated by hydrogen bonds, π - π stacking interactions, and dipole interactions among polar groups, can lead to the "collapse" of the top layer of asphaltenes. This collapse results in the cross-linking of the top layer of asphaltenes with underlying molecules and neighboring asphaltenes, forming a rigid "skin", which results in the formation of O/W emulsions with strong stability. The emulsion droplets can improve the fluidity of heavy oil when they are small-sized. Conversely, larger-sized emulsions can lead to pore throat

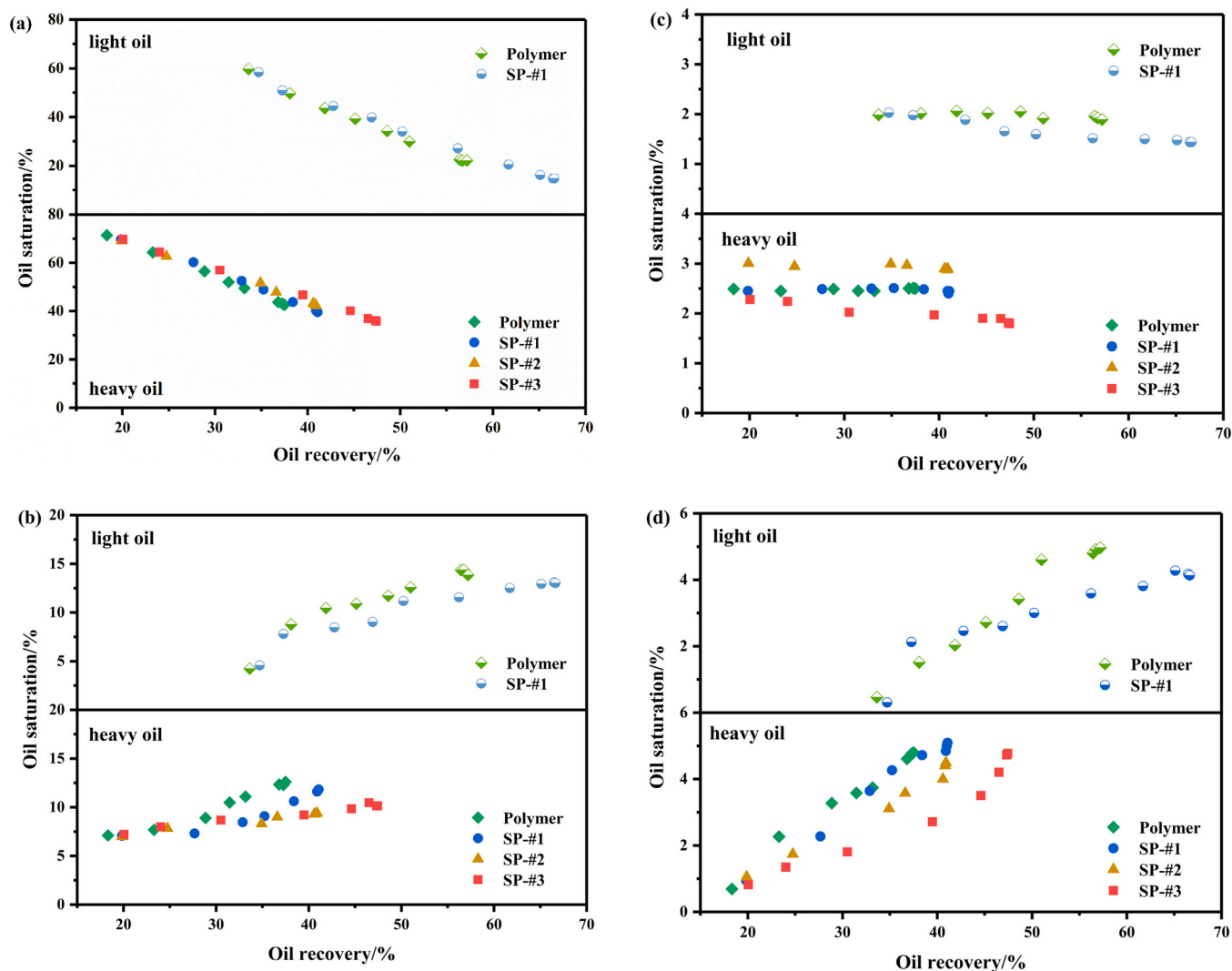


Fig. 13. Oil saturation of different types of remaining oil as a function of recovery degree with different chemical solutions using image recognition techniques. (a) microscopic heterogeneous remaining oil, (b) pore-throat residual oil, (c) hydrodynamic residual oil, and (d) film remaining oil.

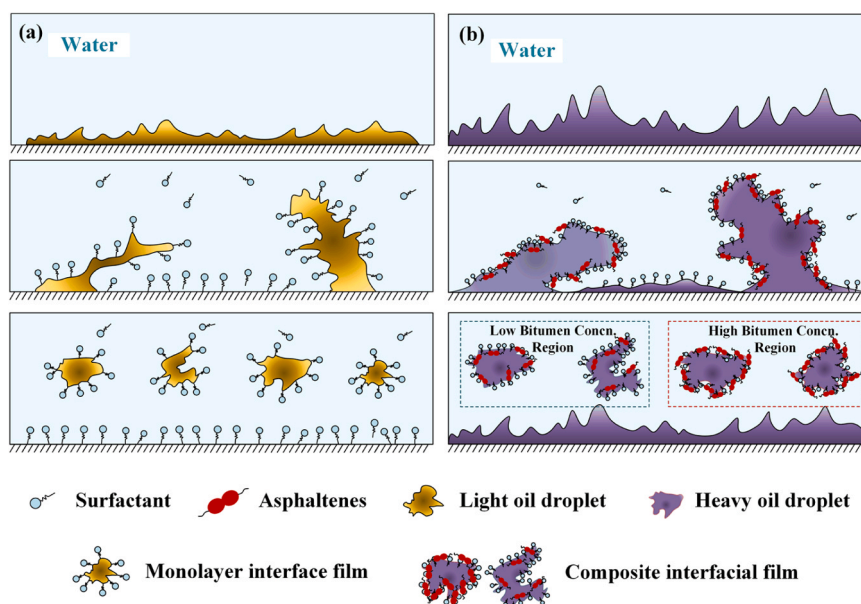


Fig. 14. Schematic diagram of the EOR mechanism of SP flooding for (a) light oil and (b) heavy oil.

blockage and an increase in swept volume [43]. However, in areas of high bitumen concentration, the interface is covered by sodium naphthenate species with cyclic and linear alkane structures, which do not allow such attraction. But asphaltenes on the interface can organize cross-link networks with active and functional groups of surfactants, forming strong "dragging" forces to drive out the remaining oil. In general, EOR by chemical flooding is the result of the combined action of heavy components and surfactant molecules in two regions. On the one hand, emulsification occurs in the low-concentration bitumen region, on the other hand, the high-strength shear viscoelasticity in the high-concentration asphalt zone has the effect of exfoliation of remaining oil.

4. Conclusion

The promising result of the micro model and core flooding tests of the SP-#3 system provide significant inspiration for implementing chemical flooding in heavy oil reservoirs. Therefore, several conclusions drawn from the above research are as follows:

- (1) The SP system with low IFT facilitates the expansion of the oil-water interface, leading to the formation of numerous oil strips that become immobilized within the porous medium due to the substantial viscous forces. This phenomenon results in much pore-throat residual oil and filmy oil.
- (2) The SP system with high interfacial viscoelasticity emphasizes the outstanding performance of exfoliation and carrying, leading to a reduction in pore-throat residual oil. While, the lack of emulsification limits the expansion of the swept volume, resulting in the occurrence of microscopic heterogeneities remaining oil increases.
- (3) The successful enhancement of heavy oil recovery by the SP system critically depends on its ability to simultaneously possess low IFT and high viscoelastic behaviors. The mere attainment of either one of these properties is insufficient to achieve optimal results.
- (4) EOR from light oil is dependent on the monolayer surfactant molecular film's ability to provide low IFT. However, heavy oil exhibits both competitive and cooperative adsorption phenomena at the oil-water interface. This phenomenon requires a

synergistic combination of viscoelasticity and emulsification to achieve high recovery.

CRediT authorship contribution statement

Han Cao: Methodology, Investigation, Experiments, Writing – original draft. **Yiqiang Li:** Conceptualization, Investigation, Writing – review & editing, Supervision. **Wenbin Gao:** Data curation, Formal analysis. **Jinxin Cao:** Visualization, Investigation. **Bingyu Sun:** Data curation, Supervision. **Jin Zhang:** Supervision.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data Availability

No data was used for the research described in the article.

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